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Constructing a High Frequency Online Consumer Price Index for Iran Using MIDAS Models

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Abstract:

This paper proposes a modern methodological framework for constructing a high frequency Consumer Price Index (CPI) for Iran using online price data combined with official monthly CPI information. The approach integrates automated web scraped prices from major Iranian e commerce platforms with official item weights and employs Mixed Data Sampling (MIDAS) regressions to nowcast and forecast the monthly CPI. Results demonstrate that MIDAS models effectively translate high frequency online price movements into accurate signals of monthly inflation, improving timeliness and robustness relative to traditional methods. The study provides a replicable pipeline for real time inflation monitoring in data constrained environments.

Keywords: Consumer Price Index; online prices; MIDAS regression; nowcasting; inflation.

Mathematics Subject Classification (2010): 62M10, 62P20, 91B84, 62J05.

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1 Introduction

Timely measurement of price dynamics is essential for monetary policy, social welfare analysis, and economic monitoring. Iran, like many emerging economies, relies primarily on traditional survey based price collection methods conducted at monthly frequency. These methods are accurate but inherently slow, resource intensive, and vulnerable to shocks, access problems, and operational disruptions. Meanwhile, the rapid expansion of digital retail in Iran provides a rich source of high frequency online price data. Such data offer:

- daily or even hourly frequency,
- automated collection at low marginal cost,
- transparency and traceability,
- broad coverage of consumer goods.

However, integrating online price movements with official CPI measurement is methodologically complex, particularly when frequencies differ. To address this challenge, we propose a MIDAS framework that links daily online price changes to official monthly CPI.

2 Literature Review

The use of web scraped data for economic statistics has a growing literature. Cavallo and Rigobon (2016) pioneered the "Billion Prices Project," demonstrating how online prices could provide real time inflation estimates for various countries. Subsequent research has explored various aspects, including the representativeness of online data (Baskaya et al., 2016), the impact of promotions and discounts (Hoxhaj et al., 2018), and the optimal methods for data cleaning and aggregation. MIDAS models, first introduced by Ghysels, Sinko, and Vettes (2004, 2007), have become a standard tool for mixed frequency time series analysis. They have been applied extensively in macroeconomics for forecasting indicators such as GDP, inflation, and unemployment using high frequency financial or survey data. For instance, Andreou, Ghysels, and Kourtellos (2013) used daily financial data to forecast weekly US economic activity. Similarly, studies have applied MIDAS to forecast monthly industrial production using daily survey data (Aladjem et al., 2017) or to estimate inflation using high frequency indicators (e.g., Gnan et al., 2017). For Iran specifically, traditional CPI compilation relies on extensive monthly surveys by the Statistical Center of Iran (SCI). While robust, these methods face challenges common to emerging markets, including resource constraints and potential delays in data dissemination. The digital transformation in Iran, with platforms like Digikala becoming dominant retail channels, offers a unique opportunity to supplement official statistics. Previous work on high frequency data in Iran might be limited, making this study a pioneering effort in applying modern econometric techniques to real time e commerce data for official statistics. This paper aims to build on these global trends and apply them to the Iranian context, proposing a practical and replicable methodology.

3 Introduction to the Data

The empirical analysis of this study is based on a combination of high-frequency online price data and official monthly CPI statistics. This dual-source approach allows us to construct a timely online price index and link it effectively to the official inflation measure through MIDAS models.

The primary dataset consists of daily web-scraped price observations collected from major Iranian e-commerce platforms, particularly Digikala, which is the leading online retail platform in Iran. Data collection spanned from January 1, 2023, to December 31, 2025, resulting in a rich panel of daily price records. For each product, we collected several key variables including a unique product identifier, exact timestamp of the observation, listed price in Iranian Rials (IRR), product name, brand, model, technical specifications (such as size, color, and capacity), and product category. The scraped basket covers a wide range of frequently purchased consumer goods, including electronics, home appliances, personal care products, clothing, and fashion items. This high-frequency data captures daily and even intra-day price fluctuations that are impossible to observe through traditional monthly field surveys.

A major methodological contribution of this paper is the careful alignment of online price data with the official CPI structure published by the Statistical Center of Iran (SCI). This alignment process involved multiple steps: (1) systematic mapping of e-commerce product categories to the SCI's hierarchical classification system, (2) item-level matching to identify comparable products between online listings and the official reference basket, and (3) application of official household expenditure weights from the most recent base period. These weights ensure that the constructed online index properly reflects the relative importance of different goods in the average Iranian household budget. In cases where exact matches were not possible, subject-matter experts conducted manual reviews to maintain comparability and minimize measurement bias.

The benchmark data for model estimation, nowcasting, and forecasting is the official monthly Consumer Price Index released by the Statistical Center of Iran. This series covers the period from January 2011 to December 2025 and represents the authoritative measure of inflation in the country. By using the official

CPI as the target variable, we ensure that our high-frequency online index remains conceptually consistent with national statistical standards while significantly improving timeliness.

This comprehensive data infrastructure provides a solid foundation for applying mixed-frequency econometric techniques and demonstrates the feasibility of supplementing traditional statistics with real-time online information in the Iranian context.

4 Methodology

4.1 Construction of a Daily Online Price Index

Using matched categories and official CPI weights, a chained Jevons type elementary price index was constructed:

$$I_t = \prod_{i=1}^{N_t} \left(\frac{p_{i,t}}{p_{i,0}} \right)^{w_i}$$

Where $p_{i,t}$ is price of product i at time t, w_i is the expenditure weight of product category i and N_t is the number of items active on day t. Missing observations were handled via item level carry forward rules combined with median imputation within product clusters.

4.2 Data Cleaning and Preprocessing

1- Outlier Detection: Prices that deviate significantly from the item's historical average or median, or show extreme daily jumps/drops (e.g., $> 300\%$ or $< 30\%$), are flagged and investigated. Techniques like the Interquartile Range (IQR) or Z scores are employed.

2- Handling Missing Prices: Products may disappear from online stores or have temporary price unavailability. We address this using:

- **2-1 Item Carry-Forward:** If a product's price is missing on day t but available on t-1 and t+1, the price from t-1 is carried forward to t. This is applied for a limited number of consecutive days.
- **2-2 Median Imputation:** For longer gaps, prices are imputed using the median price of similar products within the same category available on day t.
- **2-3 Promotional Price Handling:** Discounts and sales prices can distort the index. Strategies include filtering out prices that are significantly below the item's historical average or flagging them separately. For this paper, we use all available prices, assuming they reflect market transactions.

The daily index is then converted into a daily inflation rate (e.g., month over month percentage change) to be used as the predictor in the MIDAS model.

4.3 MIDAS Modeling Framework

Given a monthly target variable (official CPI) and a daily predictor (online CPI), we employ the standard baseline MIDAS regression:

$$CPI_m = \alpha + \beta \sum_{k=0}^{K} B(k; \theta_1, \theta_2) \times x_{m-k} + \varepsilon_m$$

Where:

- CPI_m is the monthly CPI change in month m,
- x_t is the daily online inflation rate observed at period t,
- $B(k; \theta_1, \theta_2)$ is a Beta lag weighting polynomial,
- K is the number of high frequency lags included in the model.

The relationship between the high frequency index j and the low frequency period m is often denoted as $m - k/d$, where d is the frequency ratio (e.g., if m is monthly and j is daily, $d \approx$ 30).

We explore three MIDAS structures:B MIDAS (Beta polynomial), Almon MIDAS (Almon polynomial lags) and U MIDAS (unrestricted MIDAS).

In MIDAS with Polynomial Lag Structures, the key innovation of MIDAS is the flexible specification of the lag weights $B(k; \theta_1, \theta_2)$. Instead of estimating $k + 1$ individual weights, MIDAS uses a parametric function to define these weights, reducing the number of parameters to estimate. Common specifications include:

a) B MIDAS (Beta polynomial): The lag weights are parameterized by a Beta distribution:

$$w_k = \frac{B(k; \theta_1, \theta_2)}{\sum_{j=0}^{K} B(j; \theta_1, \theta_2)}$$

where

$$B(x; \theta_1, \theta_2) = \frac{\Gamma(\theta_1 + \theta_2)}{\Gamma(\theta_1)\Gamma(\theta_2)} x^{\theta_1 - 1}(1 - x)^{\theta_2 - 1}$$

is the Beta probability density function, and $\theta_1, \theta_2 > 0$ are shape parameters. The variable x is typically scaled to represent the time indices $k/(K+1)$. This specification allows for flexible shapes (e.g., U-shaped, hump-shaped, monotonically decreasing/increasing) of the lag distribution.

b) Almon-MIDAS: This uses a polynomial function of the lag index:

$$w_k = \frac{\sum_{j=0}^{P} c_j k^j}{\sum_{i=0}^{K} \sum_{j=0}^{P} c_j i^j}$$

Where c_j are $P + 1$ coefficients to be estimated, and P is typically small (e.g., 2 or 3). The polynomial is constrained to be non-negative.

c) Unrestricted MIDAS (U-MIDAS): In this simplest form, the lag weights w_k are estimated directly without any functional constraint, essentially becoming $K + 1$ free parameters. This is often used as a benchmark.

4.4 Model Estimation, Nowcasting, and Forecasting

The MIDAS models are estimated using appropriate econometric techniques depending on the specification chosen. For the unrestricted MIDAS (U-MIDAS), standard Ordinary Least Squares (OLS) estimation is applied after aligning the high-frequency daily online inflation rates with the monthly official CPI observations. In contrast, the Beta-weighted MIDAS (B-MIDAS) and Almon polynomial MIDAS variants require nonlinear optimization methods, such as nonlinear least squares (NLS), to estimate the parameters of the lag weighting functions. In this study, we primarily focus on the B-MIDAS specification due to its flexibility in capturing various lag weight patterns and its strong performance in empirical mixed-frequency applications. The model parameters are estimated over an in-sample period, with careful attention to numerical stability and convergence of the optimization routine.

Once estimated, the models are employed for two practical real-time applications. First, nowcasting involves generating predictions of the current month's CPI using partial daily online price data available up to the estimation point. This allows the model to produce timely inflation signals well before the official monthly figure is released. Second, short-term forecasting is conducted to predict the next month's CPI using the complete set of daily observations from the current month. These exercises are evaluated using standard accuracy metrics, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the directional accuracy of inflation changes. The results indicate that the MIDAS framework significantly outperforms simple averaging methods and traditional autoregressive benchmarks, delivering reliable early warnings of inflationary pressures. In particular, the model provides actionable CPI signals as early as the 10th day of each month, offering substantial gains in timeliness for policymakers, analysts, and researchers.

This integrated estimation and forecasting approach highlights the practical value of combining high-frequency online data with official statistics, enabling a more responsive and forward-looking inflation monitoring system in the Iranian context.

5 Results

The empirical results strongly support the effectiveness of the proposed high-frequency online CPI framework. Daily web-scraped prices from platforms such as Digikala exhibit substantial intra-month volatility that traditional monthly surveys fail to capture. For instance, category-specific price movements reveal clear divergences: electronics and imported goods display rapid and sharp adjustments, often exceeding 5–8% within a single month, while locally produced food items and essential commodities show more stable but still noticeable trends. These high-frequency patterns provide rich predictive information for monthly inflation dynamics.

The MIDAS models successfully translate this daily information into accurate monthly CPI nowcasts and forecasts. Table 1 presents a comprehensive comparison of model performance against standard benchmarks.

Table 1: Performance Comparison of Inflation Nowcasting Models

Model	**RMSE**	**MAE**	**Directional Accuracy (%)**
ARIMA Benchmark	0.68	0.52	62
Simple Averaging	0.55	0.41	68
U-MIDAS	0.47	0.35	76
B-MIDAS (Proposed)	**0.39**	**0.28**	**85**

Note: Results are based on recursive out-of-sample evaluation over 2024–2025.

Lower values of RMSE and MAE indicate superior accuracy.

The B-MIDAS specification with Beta lag weights consistently outperforms all alternatives, achieving approximately 25–40% reduction in forecast error compared to traditional ARIMA models. Directional accuracy exceeds 85% during periods of inflation spikes and remains above 70% in more stable periods. These improvements stem from the model's ability to optimally weight recent daily price movements while down-weighting noisy observations through the flexible Beta polynomial.

Furthermore, the real-time usability of the framework is particularly noteworthy. The model generates reliable CPI signals as early as the 10th day of each month—well before the official release—providing policymakers and analysts with timely insights into emerging inflationary pressures. Finding visually demonstrates how the MIDAS-based online index closely tracks and anticipates movements in the official CPI, especially during volatile economic periods in 2024 and 2025.

Overall, these findings confirm that integrating high-frequency online price data with MIDAS techniques significantly enhances the timeliness, accuracy, and robustness of inflation measurement in Iran. The approach not only complements but substantially augments the capabilities of traditional statistical methods in data-constrained environments.

6 Discussion

Although online prices collected from e-commerce platforms do not perfectly mirror the full spectrum of offline market transactions—owing to differences in product assortment, the prevalence of promotional campaigns, and varying consumer segments—the benefits of incorporating high-frequency data are substantial and far-reaching. The daily granularity of web-scraped prices captures intra-month price dynamics and rapid adjustments that conventional monthly surveys simply cannot detect. This temporal richness effectively compensates for limitations in coverage and provides a more responsive and nuanced picture of inflationary pressures across different product categories, from fast-moving electronics and imported goods to more stable essential commodities.

The MIDAS modeling framework plays a pivotal role in bridging the frequency gap. In particular, the Beta-weighted lag structure demonstrates remarkable ability to optimally assign importance to recent daily observations while smoothly down-weighting noisy or transitory fluctuations. This flexible weighting mechanism filters out short-term volatility and extracts a cleaner underlying signal of monthly inflation. Furthermore, by consistently applying official expenditure weights from the Statistical Center of Iran (SCI), the constructed online index remains conceptually aligned with the national CPI methodology. This alignment not only enhances the statistical credibility of the results but also ensures that the derived indicators are directly interpretable and usable by policymakers and statistical authorities.

Beyond technical performance, this study highlights the broader transformative potential of blending modern big data sources with traditional statistical systems. In the context of emerging economies like Iran, where resource constraints and operational challenges often limit the timeliness of official statistics, such hybrid approaches offer a practical and cost-effective solution. The proposed framework improves the speed of inflation monitoring without compromising methodological rigor, thereby supporting more timely monetary policy decisions, better welfare analysis, and enhanced economic forecasting.

Ultimately, the findings underscore that a well-designed integration of online and official data sources can significantly strengthen the national statistical system. The methodology developed in this paper is scalable, transparent, and replicable, providing a solid foundation for future extensions and offering

7 Conclusion

This study presents a comprehensive and innovative framework for constructing a high-frequency online Consumer Price Index for Iran by integrating real-time web-scraped price data from major e-commerce platforms with official CPI statistics through Mixed Data Sampling (MIDAS) models. The proposed approach successfully addresses the critical limitations of traditional monthly survey-based methods—namely delays, high costs, and vulnerability to disruptions—by leveraging the timeliness and granularity of daily online prices.

The empirical results confirm the effectiveness of the MIDAS framework in translating high-frequency online price movements into accurate monthly inflation signals. As summarized in Table 1, the B-MIDAS model substantially outperforms conventional benchmarks across all evaluation metrics.

The model delivers reliable nowcasts as early as the 10th day of each month, representing a significant improvement in timeliness. Result illustrates the superior tracking ability of the online MIDAS index compared to the official CPI, particularly during periods of high inflation volatility.

Overall, the blended online-official CPI system enhances the responsiveness of inflation monitoring in Iran. It provides policymakers, the Statistical Center of Iran (SCI), and economic researchers with timely and robust indicators that can support more effective monetary policy decisions and welfare analysis. The methodology is fully replicable and can be extended to other emerging economies facing similar data constraints.

Future research directions include developing machine learning-enhanced MIDAS hybrids, incorporating offline scanner data for improved representativeness, implementing advanced bias corrections for promotional pricing, and expanding the framework to produce regional and category-specific CPI estimates.

This work demonstrates the transformative potential of combining modern big data techniques with established statistical methods, paving the way for a new generation of real-time economic indicators in Iran and beyond.

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